Exploratory Data Analysis: Lung Cancer Dataset

Assignment 3

1. Dataset Overview

This project explores a cleaned lung cancer dataset to understand the distribution of patient, clinical, treatment, and laboratory variables and how they relate to the target, Survival_Group (Short-Term or Long-Term). The saved notebook reports 23,658 rows and 48 columns before the five engineered features were added. The analysis includes descriptive statistics, visualizations, correlation analysis, feature engineering, Variance Threshold, and PCA.

Target distribution:

Survival Group	Patients	Percentage
Long-Term	11,948	50.50%
Short-Term	11,710	49.50%

Descriptive Statistics

The average patient age was about 54.4 years (standard deviation 14.4). Average tumor size was about 55.4 mm (standard deviation 26.0). Mean systolic and diastolic blood pressure were approximately 134.5 and 84.5, respectively. These summaries show variation across the patient population and provide context for the visual analysis.

2. Correlation Findings and Business Insights

The Pearson correlation analysis found no predictor pairs above the notebook's selected absolute correlation threshold of 0.70. The strongest listed correlations were mostly between one-hot encoded categories (for example, different tumor-location or smoking-history categories), which are expected to be negatively related when categories are mutually exclusive. Individual numeric predictors had very small correlations with the survival target; the largest displayed target correlation was about 0.013 in absolute value. Correlation alone therefore did not identify a strong single predictor of survival.

Five business insights supported by the visual analysis

1. Balanced outcome: Long-Term survival represents 50.50% and Short-Term survival 49.50% of the dataset, so there is no major class imbalance.

2. Age: The mean age is approximately 54.5 years for Long-Term patients and 54.4 years for Short-Term patients. The overlapping distributions suggest age alone may not separate the groups well.

3. Tumor size: Mean tumor size is approximately 55.6 mm for Long-Term patients and 55.1 mm for Short-Term patients. The small difference and overlapping distributions suggest limited separation by tumor size alone.

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4. Cancer stage: The stage count plot shows broadly similar outcome distributions across stages in this dataset; stage alone does not create a strong visual separation in the observed data.

5. Weak individual correlations: Numeric predictor correlations with the encoded survival outcome are small, suggesting that a useful model may need to combine multiple variables and capture patterns not represented by simple linear correlations.

3. Engineered Features

Feature	How it was created	Why it was added
Age_Group	Age grouped into 30–49, 50–64, and 65–79 bands.	To compare survival patterns across broader age ranges.
Tumor_Size_Group	Tumor size grouped as Small (up to 40 mm), Medium (40–70 mm), and Large (over 70 mm).	To make tumor-size patterns easier to compare categorically.
Comorbidity_Count	Sum of seven yes/no comorbidity indicators.	To summarize the number of recorded comorbidities for each patient.
Treatment_Count	Sum of three treatment indicator columns.	To create a compact count from the treatment indicators; interpretation depends on how those indicators are encoded.
Blood_Pressure_Difference	Systolic blood pressure minus diastolic blood pressure.	To represent the difference between the two blood-pressure measurements.

4. Variance Threshold, PCA, and Modeling Reflection

Variance Threshold

The notebook applied Variance Threshold with a threshold of 0.01. The recorded output retained all 50 predictors, so this filter did not reduce the feature count. This suggests that none of the included predictors fell below the selected variance threshold.

Principal Component Analysis (PCA)

In the saved notebook outputs, PCA required 39 components to retain 90% of the variance and 42 components to retain 95% of the variance, compared with 50 original predictors. At the 90% level, this is 11 fewer components, or a 22% reduction in feature count.

Original features versus PCA features

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Approach	Features	Logistic Regression accuracy
Original Features	50	49.79%
PCA Features (90% variance)	39	49.64%

The recorded accuracies are very similar: approximately 49.79% using the original features and 49.64% using PCA features. These results are close to chance-level accuracy for a nearly balanced two-class target, so this simple Logistic Regression model should not be presented as a strong predictor. PCA reduced dimensionality with only a small difference in accuracy, but principal components are harder to interpret than the original variables.

Reflection

EDA helped me understand the structure of the dataset before modeling. The plots and correlation analysis showed that the survival groups were balanced and that individual variables did not strongly separate the outcomes. This encouraged me to engineer grouped and summary features and to test a dimensionality-reduction method instead of assuming that one variable would explain survival.

Conclusion

Overall, the analysis provided a clearer view of patient characteristics, survival-group distributions, and relationships among predictors. The five engineered features offered additional ways to examine the data. Variance Threshold did not remove features, while PCA reduced the feature count from 50 to 39 at 90% explained variance. The similar Logistic Regression results suggest that further modeling, validation, and feature investigation are needed before drawing predictive conclusions.